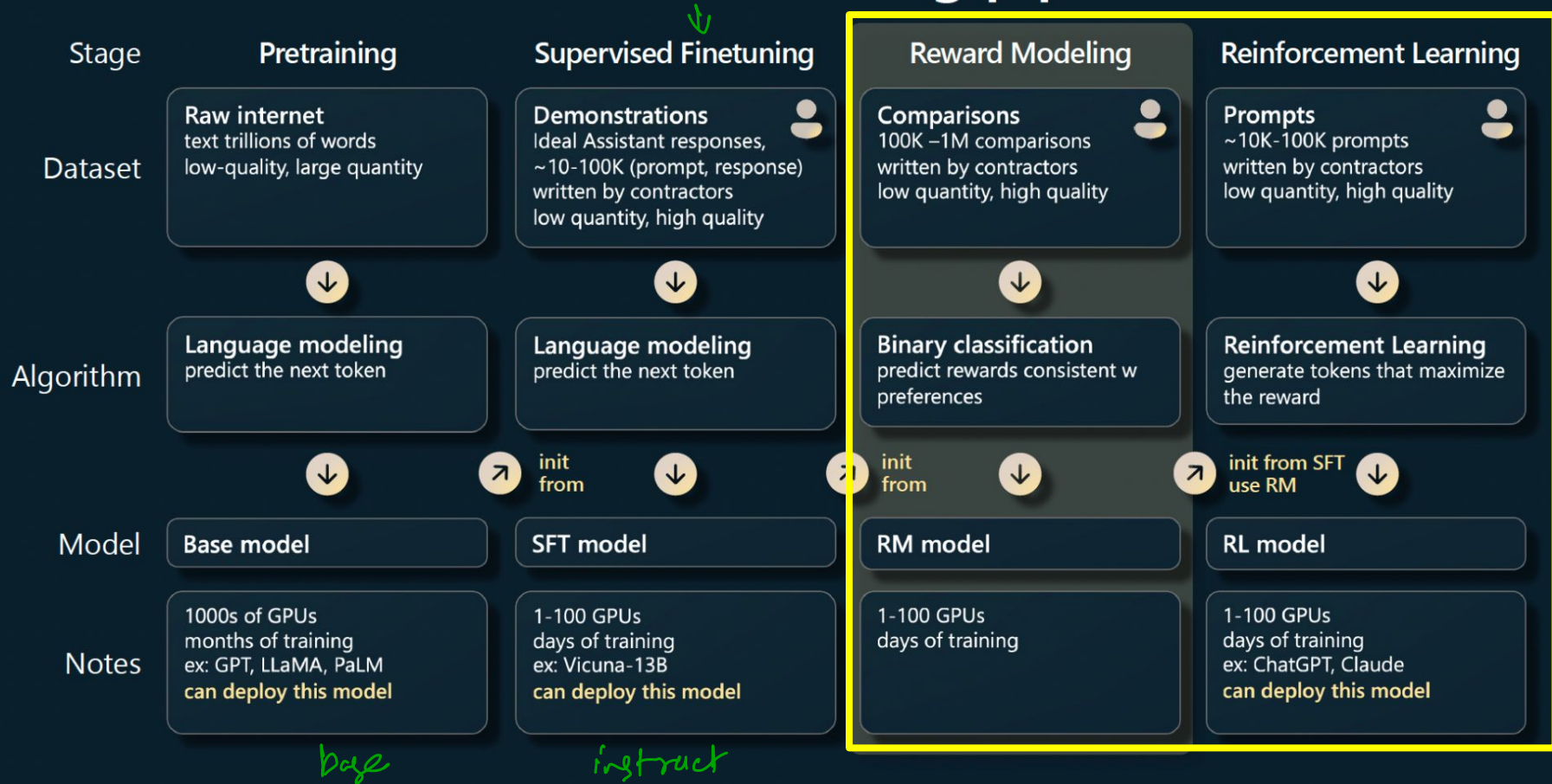


Reinforcement Learning w/ Human Feedback

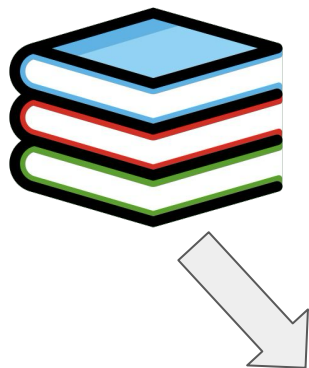
CS60216: Safety Fundamentals for Generative AI

Jan 21st, 2026

GPT Assistant training pipeline



History: early OpenAI experiments with RLHF



“Three pigs defend themselves from a mean wolf”

1 Collect human feedback

A Reddit post is sampled from the Reddit TL;DR dataset.



Various policies are used to sample a set of summaries.



Two summaries are selected for evaluation.



A human judges which is a better summary of the post.



“j is better than k”

2 Train reward model

One post with two summaries judged by a human are fed to the reward model.



The reward model calculates a reward r for each summary.



The loss is calculated based on the rewards and human label, and is used to update the reward model.

$$\text{loss} = \log(\sigma(r_j - r_k))$$

“j is better than k”

3 Train policy with PPO

A new post is sampled from the dataset.



The policy π generates a summary for the post.



The reward model calculates a reward for the summary.



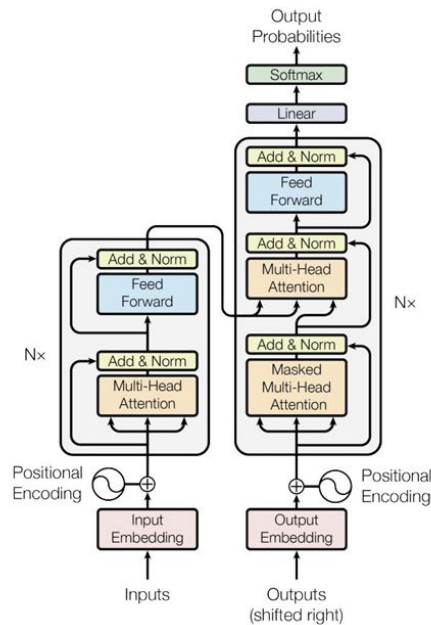
The reward is used to update the policy via PPO.



r

Figure 2: Diagram of our human feedback, reward model training, and policy training procedure.

Three phases of RLHF

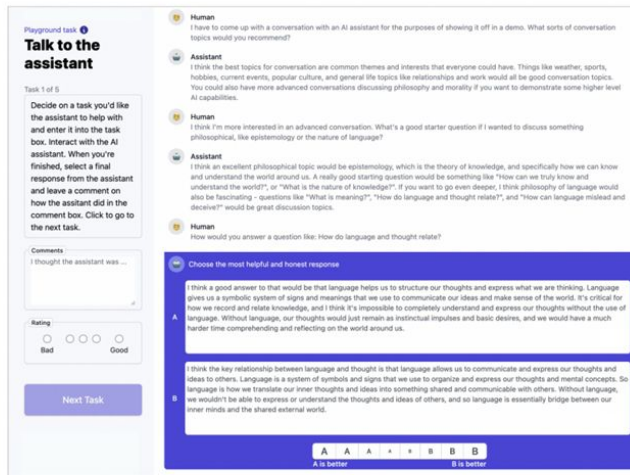


Vaswani et al. 2017

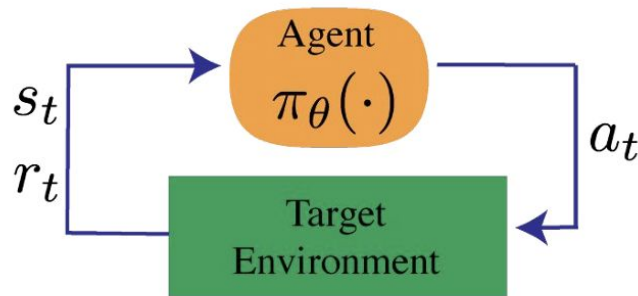
Instruction.

base model (instruction, helpful, chatty etc.)

preference collection & training



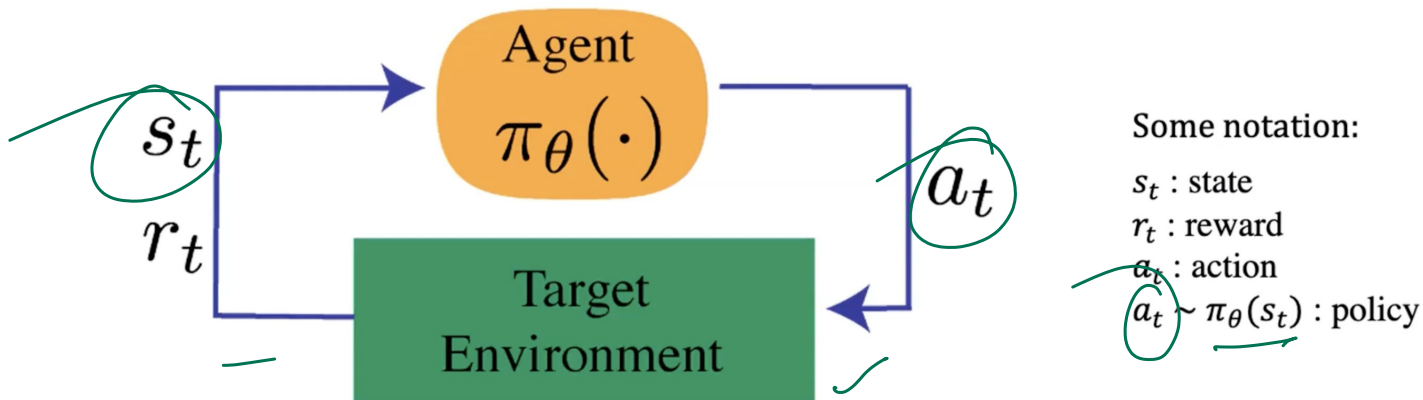
reinforcement learning optimization



We already talked about instruction fine-tuning

Reinforcement Learning: The Basics

- An agent **interacts** with an environment by taking **actions**
- The environment returns a **reward** for the **action** and a **new state** (representation of the world at that moment).
- Agent uses a **policy** function to choose an action at a given **state**.
- Quite an open-ended learning paradigm.



Reinforcement Learning: Definitions

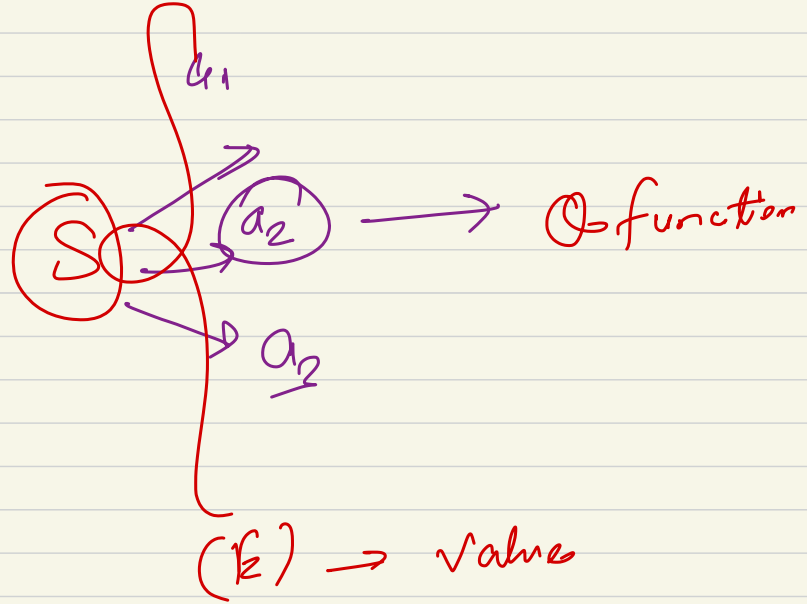
- **Reward (r):** A scalar value indicating the desirability of an action or state, typically denoted as r . ✓
- **Action (a):** A decision or move made by an agent in an environment, often represented as $a \in A$, where A is the set of possible actions.
- **State (s):** The current configuration or situation of the environment, usually denoted as $s \in S$, where S is the state space.
- **Trajectory (τ):** A trajectory is a sequence of states, actions, and rewards experienced by an agent: $\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_T, a_T, r_T)$.
- **Trajectory Distribution ($(\tau|\pi)$):** The probability of a trajectory under policy π is $P(\tau|\pi) = p(s_0) \prod_{t=0}^T \pi(a_t|s_t)p(s_{t+1}|s_t, a_t)$, where $p(s_0)$ is the initial state distribution and $p(s_{t+1}|s_t, a_t)$ is the transition probability. ✓

$$p(s_0) \quad \pi(a_0 | s_0) \quad p(s_1 | s_0, a_0)$$

Reinforcement Learning: Definitions

- **Policy (π):** In RL, a policy is a strategy or rule that the agent follows to decide which action to take in a given state: $\pi(a|s)$.
- **Value Function (V):** A function that estimates the expected cumulative reward from a given state: $V(s) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r_t | s_0 = s]$.
- **Q-Function (Q):** A function that estimates the expected cumulative reward from taking a specific action in a given state: $Q(s, a) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r_t | s_0 = s, a_0 = a]$.
- **Advantage Function (A):** The advantage function $A(s, a)$ quantifies the relative benefit of taking action a in state s compared to the average action. It's defined as $A(s, a) = Q(s, a) - V(s)$. Advantage functions (and value functions) can depend on a specific policy, $A^\pi(s, a)$.

Q-V



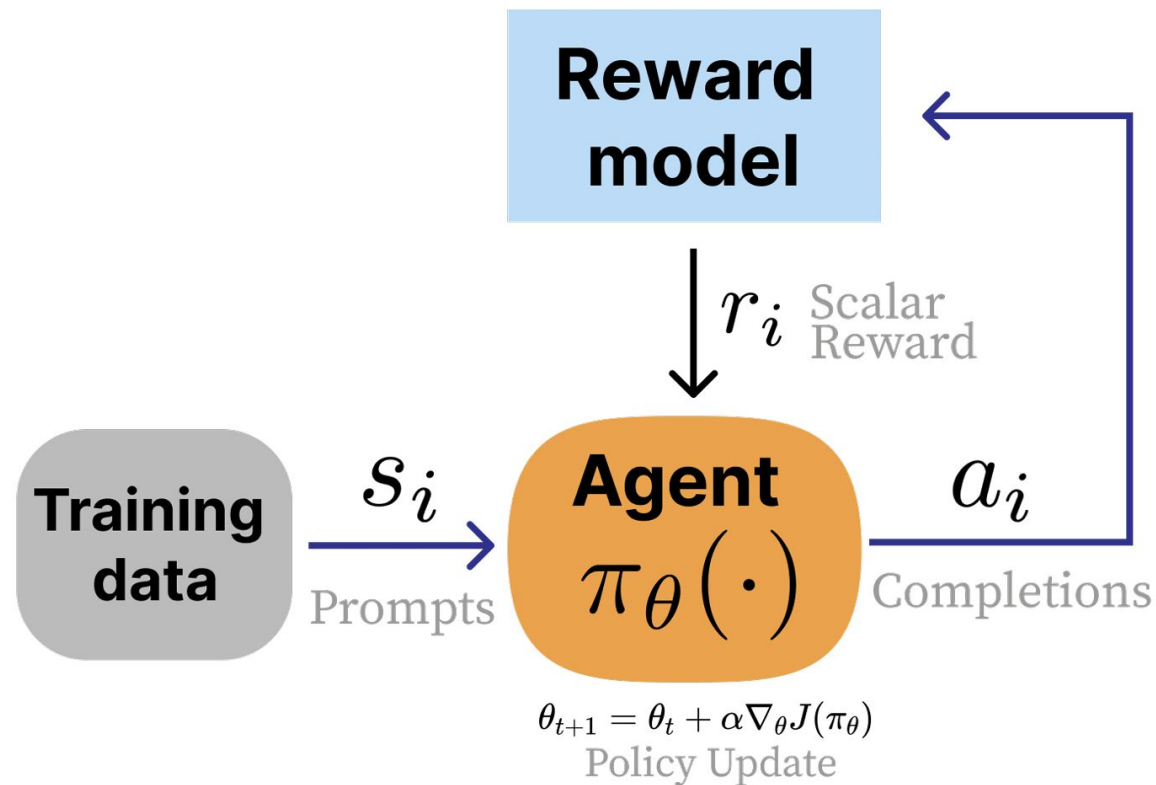
Reinforcement Learning: Definitions

- **Expectation of Reward Optimization:** The primary goal in RL, which involves maximizing the expected cumulative reward:

$$\max_{\theta} \mathbb{E}_{s \sim \rho_{\pi}, a \sim \pi_{\theta}} [\sum_{t=0}^{\infty} \gamma^t r_t]$$

where ρ_{π} is the state distribution under policy π , and γ is the discount factor.

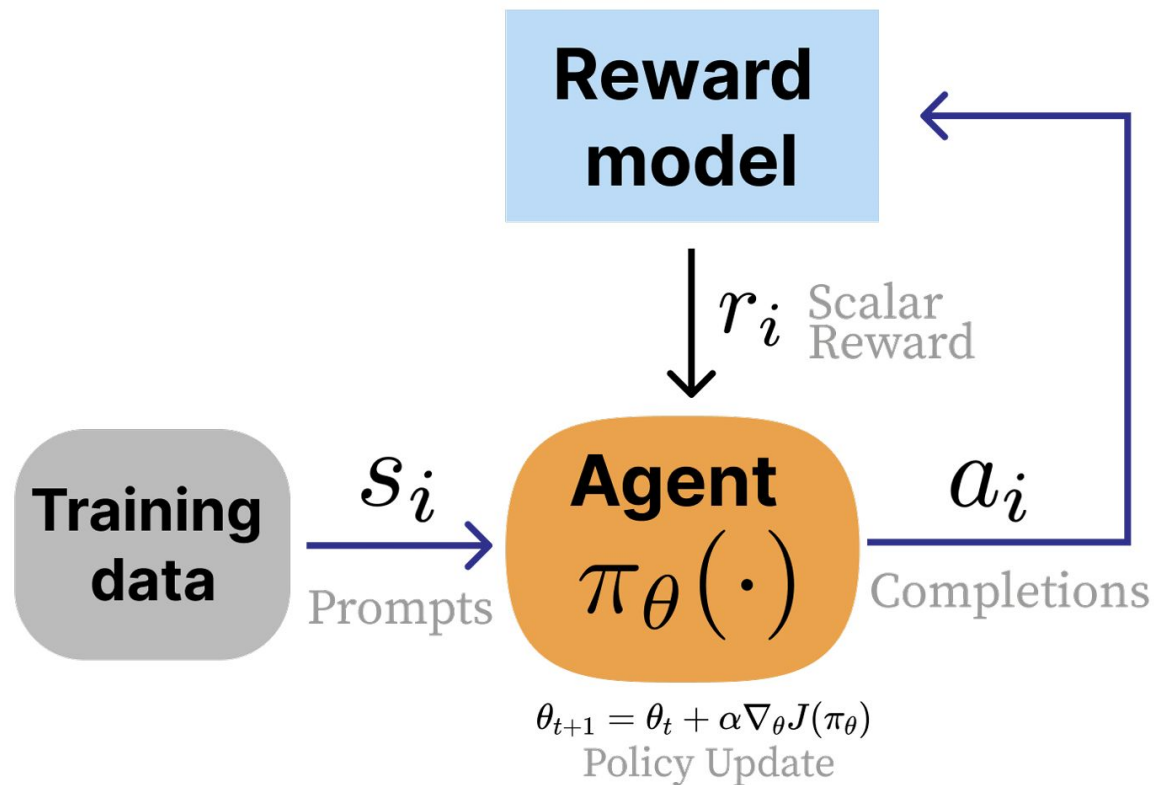
Standard RLHF loop: How is it different?



No state transitions exist.

In RLHF, the initial states for the domain are prompts sampled from a training dataset and the “action” is the completion to said prompt. During standard practices, this action does not impact the next state and is only scored by the reward model.

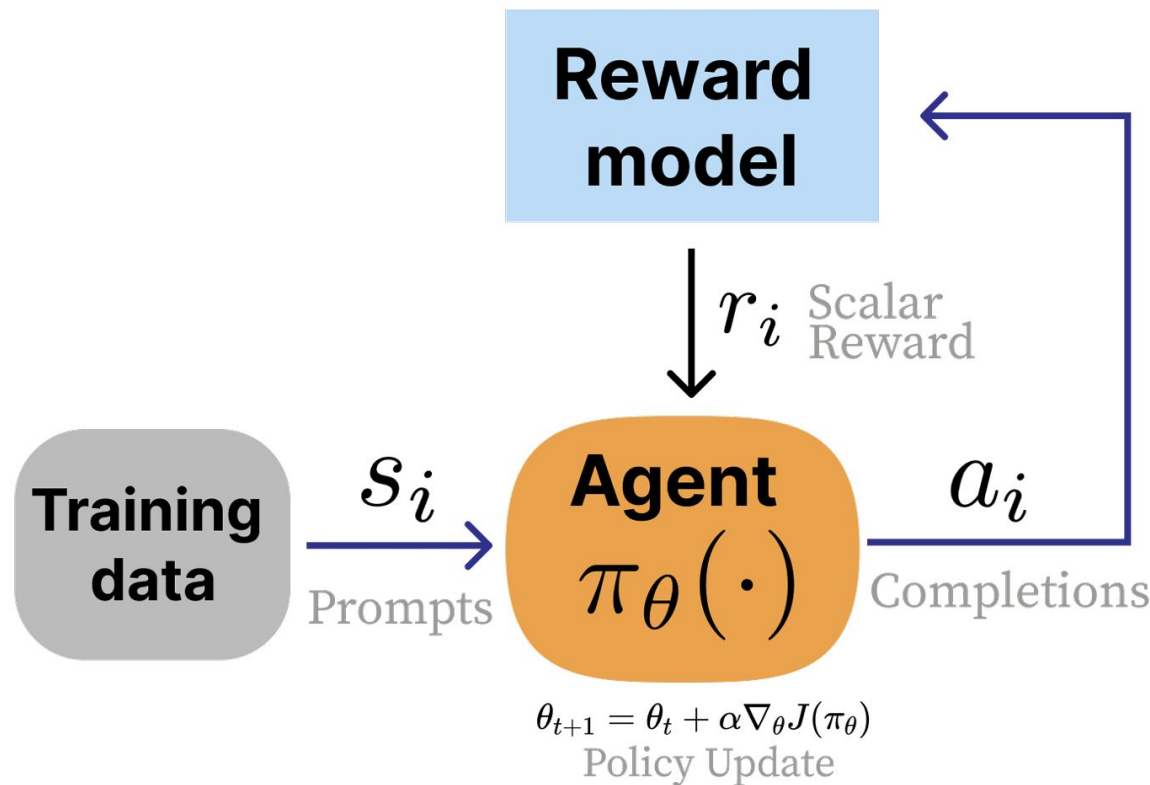
Standard RLHF loop: How is it different?



Response level rewards.

RLHF attribution of reward is done for an entire sequence of actions, composed of multiple generated tokens, rather than in a fine-grained manner.

Standard RLHF loop: How is it different?



Switching from a reward function to a reward model.

In RLHF, a learned model of human preferences, $r_{\theta}(s_t, a_t)$ is used instead of an environmental reward function. This gives the designer a substantial increase in the flexibility of the approach and control over the final results.

Why the reward model?

Problem 1: human-in-the-loop is expensive! ✓

- **Solution:** instead of directly asking humans for preferences, **model their preferences** as a separate (NLP) problem! [[Knox and Stone, 2009](#)]

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

$$s_1$$
$$R(s_1) = 8.0$$


The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

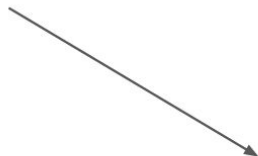
$$s_2$$
$$R(s_2) = 1.2$$


Train an LM $RM_\phi(s)$ to
predict human
preferences from an
annotated dataset, then
optimize for RM_ϕ instead.

Feedback interface

Preference collection

LLM provides two options for next responses



- Human**
I have to come up with a conversation with an AI assistant for the purposes of showing it off in a demo. What sorts of conversation topics would you recommend?
- Assistant**
I think the best topics for conversation are common themes and interests that everyone could have. Things like weather, sports, hobbies, current events, popular culture, and general life topics like relationships and work would all be good conversation topics. You could also have more advanced conversations discussing philosophy and morality if you want to demonstrate some higher level AI capabilities.
- Human**
I think I'm more interested in an advanced conversation. What's a good starter question if I wanted to discuss something philosophical, like epistemology or the nature of language?
- Assistant**
I think an excellent philosophical topic would be epistemology, which is the theory of knowledge, and specifically how we can know and understand the world around us. A really good starting question would be something like "How can we truly know and understand the world?", or "What is the nature of knowledge?". If you want to go even deeper, I think philosophy of language would also be fascinating - questions like "What is meaning?", "How do language and thought relate?", and "How can language mislead and deceive?" would be great discussion topics.
- Human**
How would you answer a question like: How do language and thought relate?

Choose the most helpful and honest response

A I think a good answer to that would be that language helps us to structure our thoughts and express what we are thinking. Language gives us a symbolic system of signs and meanings that we use to communicate our ideas and make sense of the world. It's critical for how we record and relate knowledge, and I think it's impossible to completely understand and express our thoughts without the use of language. Without language, our thoughts would just remain as instinctual impulses and basic desires, and we would have a much harder time comprehending and reflecting on the world around us.

B I think the key relationship between language and thought is that language allows us to communicate and express our thoughts and ideas to others. Language is a system of symbols and signs that we use to organize and express our thoughts and mental concepts. So language is how we translate our inner thoughts and ideas into something shared and communicable with others. Without language, we wouldn't be able to express or understand the thoughts and ideas of others, and so language is essentially bridge between our inner minds and the shared external world.

A A A A B B B

A is better B is better



Hugging Face



Toloka

RLHF at ICML 2023, 32

Feedback interface

-  **Human**
I have to come up with a conversation with an AI assistant for the purposes of showing it off in a demo. What sorts of conversation topics would you recommend?
-  **Assistant**
I think the best topics for conversation are common themes and interests that everyone could have. Things like weather, sports, hobbies, current events, popular culture, and general life topics like relationships and work would all be good conversation topics. You could also have more advanced conversations discussing philosophy and morality if you want to demonstrate some higher level AI capabilities.
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How would you answer a question like: How do language and thought relate?

human rates better response

 Choose the most helpful and honest response

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☒ A ☐ A ☐ A ☐ A ☐ B ☐ B ☐ B ☐ B

A is better B is better



Hugging Face



Why pairwise?

- **Problem 2:** human judgments are noisy and miscalibrated!
- **Solution:** instead of asking for direct ratings, ask for **pairwise comparisons**, which can be more reliable [[Phelps et al., 2015](#); [Clark et al., 2018](#)]

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

s_3

$$R(s_3) = 4.1? \quad 6.6? \quad 3.2?$$

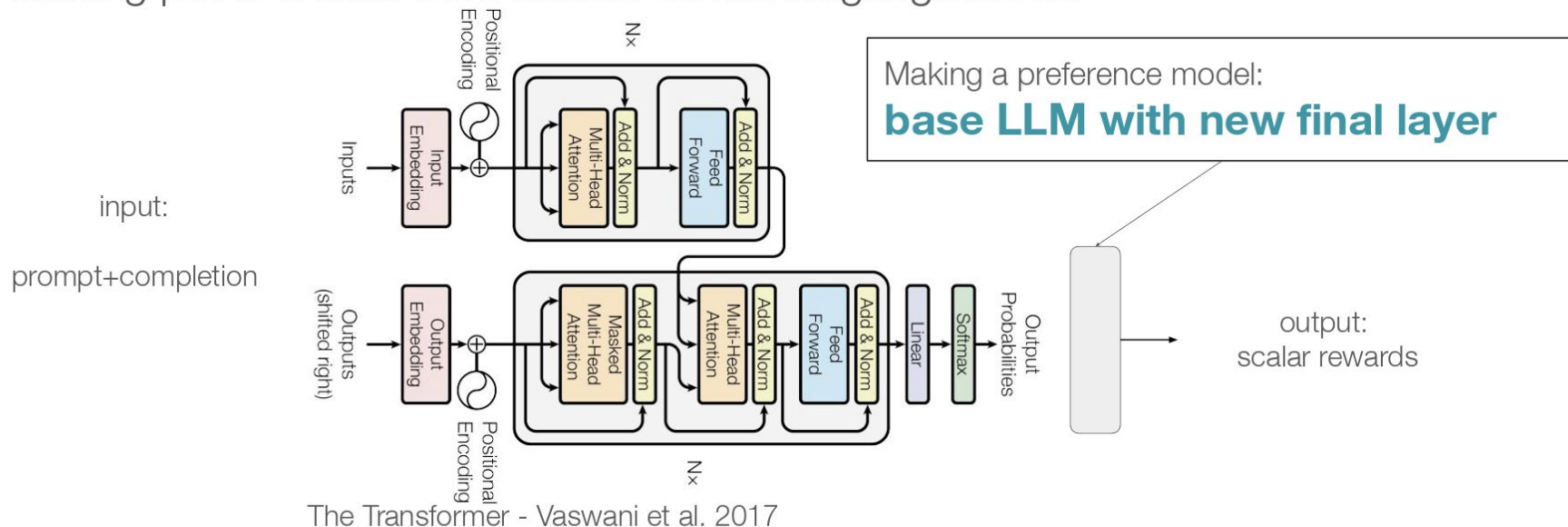
Preference Model: Pairwise comparison

Question (Prompt)	Answer 1	Answer 2	Chosen
Where is Shanghai?	Shanghai is a city in China ✓	Shanghai does not exist ✓	1 ✓
Explain gravity like I'm 5	Gravity is a famous restaurant	Gravity is what pulls things toward each other. It's why you stay on the ground and planets orbit the sun.	2
What is 2+2?	4	2+2 is a very complicated math problem...	1

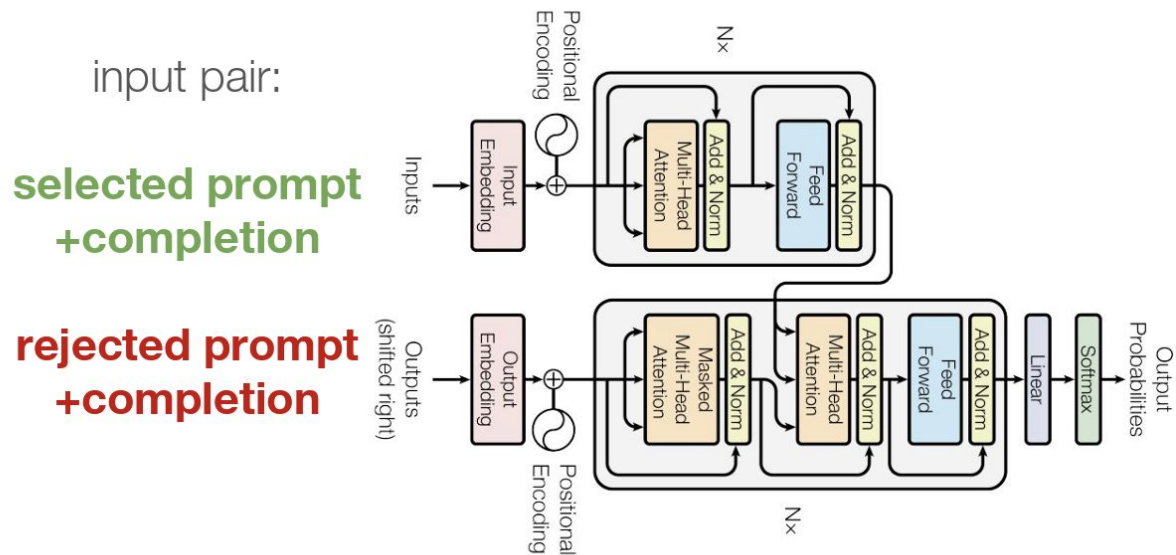
Preference model structure

Also known as the reward model

starting point: a base **instruction-tuned** language model



Preference model structure



The Transformer - Vaswani et al. 2017

r_p

r_w

outputs:
two scalar rewards

What is an appropriate loss?



Hugging Face



Training the reward model

Bradley-Terry [1952] paired comparison model

$$J_{RM}(\phi) = -\mathbb{E}_{(s^w, s^l) \sim D} [\log \sigma(RM_{\phi}(s^w) - RM_{\phi}(s^l))]$$

“winning”
sample

“losing”
sample

s^w should score
higher than s^l

$$s^w > s^l$$



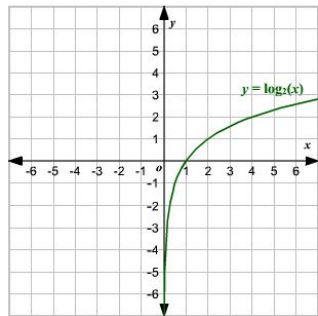
~~lost~~
small the value

$$s^l > s^w$$



large the value

Training the reward model

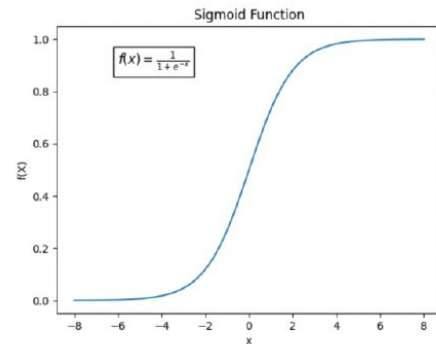


$$Loss = -\log \sigma(r(x, y_w) - r(x, y_l))$$

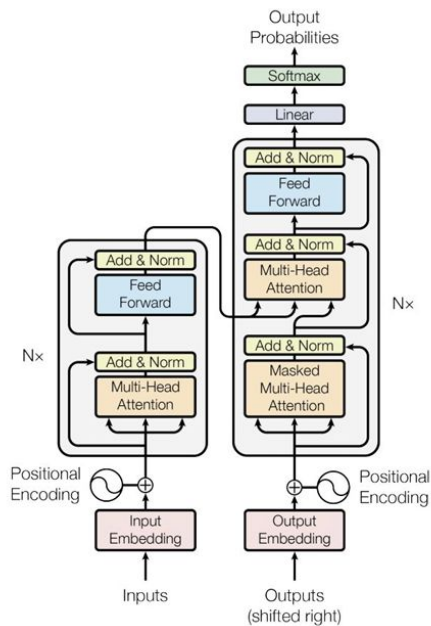
We have two cases:

- $r(x, y_w) > r(x, y_l) \rightarrow$ Sigmoid will return a value $> 0.5 \rightarrow$ Loss will be a small negative number
 - The loss will be small when the order is correct
- $r(x, y_w) < r(x, y_l) \rightarrow$ Sigmoid will return a value $< 0.5 \rightarrow$ Loss will be a very large negative number
 - The loss will be very large when the order is wrong

This loss forces the model to give high rewards to “winning” responses and low rewards to “losing” responses, because that’s the only way for the model to minimize the loss.

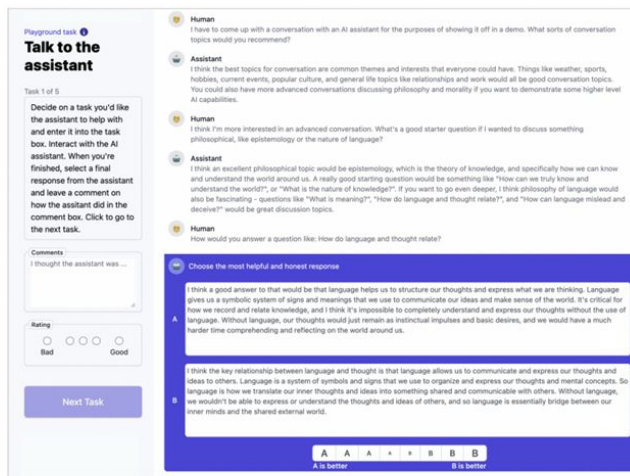


Three phases of RLHF

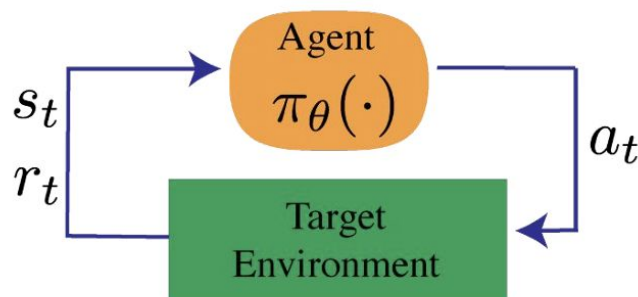


Vaswani et al. 2017

preference collection & training



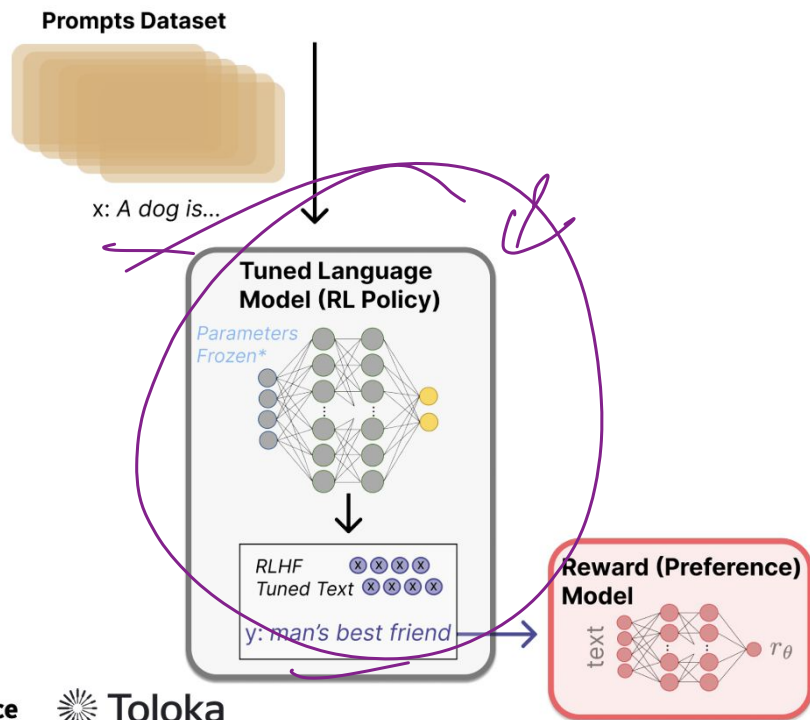
reinforcement learning optimization



base model (instruction, helpful, chatty etc.)

We already talked about instruction fine-tuning and preference model

Fine tuning with RL - using a reward model



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Toloka

RLHF Objective

- How do we actually change our LM parameters θ to maximize this?

$$\mathbb{E}_{\hat{s} \sim p_{\theta}(s)}[R(\hat{s})]$$

- Let's try doing gradient ascent!

$$\theta_{t+1} := \theta_t + \alpha \nabla_{\theta_t} \mathbb{E}_{\hat{s} \sim p_{\theta_t}(s)}[R(\hat{s})]$$

How do we estimate
this expectation??



What if our reward
function is non-
differentiable??

